

DAVID ABRIL
QA Automation Engineer
Email: davidstevenabril@gmail.com
Portfolio: <https://www.daveautomation.dev/>
LinkedIn: <https://www.linkedin.com/in/david-abril-189099138>
English: C2 (EF SET Certified)

PROFESSIONAL SUMMARY

QA Automation Engineer with backend development foundations and experience designing scalable test automation frameworks, CI/CD-integrated pipelines, and secure validation systems.

Proven record of reducing deployment timelines by 43%, eliminating 140,000+ manual DOM interactions, and saving 150+ engineering hours through automation architecture.

Experienced in Playwright, Cypress, Puppeteer, API testing, Azure DevOps CI/CD, regression strategy, and cross-functional delivery in Agile environments.

Strong background in predictive forecasting, operational optimization, and secure backend data handling.

TECHNICAL SKILLS

Test Automation:

Playwright, Cypress, Puppeteer, Selenium concepts, POM, BDD/Gherkin, UI Testing, End-to-End/Backend Testing, Cross-Browser, Performance, Accessibility & Security.

API & Backend Testing:

REST APIs, GraphQL, Postman, SQL, MySQL, MongoDB, JSON validation, HTTP status codes, Headers validation.

Programming:

JavaScript, TypeScript, Java, C#, SQL, PostgreSQL, HTML, CSS, React, Node.js, Gherkin, PHP.

DevOps & CI/CD:

Azure DevOps (YAML Pipelines), Git branching strategies, Pull Requests, Docker, Wrangler, Artifact Packaging, Continuous Integration, Continuous Delivery, Automated Regression Pipelines.

Testing Strategy:

Regression Testing, Smoke, Integration, Functional, Security, Accessibility, API Performance, SHA-256 hashing, OpenAI Embeddings Testing, Scheduled/Nightly runs, LLM/AI Testing.

Tools & Reporting:

Allure Reporting, Debug Tracing, Video Reporting, Logging, Test Fixtures, Error Interception, K6 performance observability, Jira, Artifact generation & automatic github report summary generation per run.

PROFESSIONAL EXPERIENCE

Website Migration Team Lead & Automation

Nationwide Marketing Group | 2025–2026 |

Remote (US-based organization)

- Lead a 9-person cross-functional team migrating e-commerce platforms from legacy systems to modern secure architecture.
- Built a data-driven capacity forecasting model influencing hiring and resource allocation decisions.
- Designed and deployed a custom Playwright automation framework automating 11 critical site setup workflows.
- Eliminated 113 manual steps per site and removed 140,000+ manual DOM interactions.

- Saved 150+ engineering hours immediately across backlog of 300+ sites.
- Integrated automation framework into Azure DevOps CI/CD pipelines.
- Standardized bug escalation and release validation procedures.
- Conduct regression, smoke, and integration testing across multiple deployments.

E-Commerce Migration Engineer

Nationwide Marketing Group | 2024–2025 | Remote

- Migrated legacy e-commerce platforms using advanced HTML, CSS, and JavaScript into responsive modern systems.
- Achieved highest migration conversion rate within team.
- Improved SEO baselines and mobile responsiveness across client sites.
- Proposed workflow improvements adopted across multiple departments.

Backend Software Developer

Revista Credencial | 2014 | Bogotá, Colombia

- Developed secure Java + SQL data transformation system handling sensitive banking information.
- Implemented 3-layer security architecture with role-based access controls.
- Built decrypt → transform → re-encrypt processing pipeline.
- Executed database-level queries for secure data validation.
- Ensured compliance and secure handling of financial data.

AUTOMATION & ENGINEERING PROJECTS

AI-Powered Portfolio Chatbot (RAG Architecture + Secure CI/CD)

Designed and implemented a production-grade AI chatbot integrated into personal

portfolio to create an interactive hiring experience.

Technical Scope:

- Retrieval-Augmented Generation (RAG) architecture
- Semantic chunking & vector similarity search
- Embedding-based retrieval with threshold tuning
- Secure backend worker implementation
- Prompt injection protection
- Context builder with guardrails

Testing & Quality Engineering:

- Unit tests for context builder and security guard
- Injection and prompt attack validation tests
- Smoke tests validating deployment artifacts
- Automated CI/CD pipeline execution
- LLM retrieval accuracy
- Scheduled nightly regression testing module
- End-to-end UI testing (Firefox, Chromium & Webkit) with artifact verification

Impact:

- Demonstrates applied AI engineering + QA rigor
- Converts static portfolio into interactive system
- Showcases automation maturity beyond UI testing
- Validates full lifecycle ownership (architecture → testing → CI/CD → deployment)

Azure DevOps CI/CD Full Lifecycle Project

- Built and deployed React application through complete DevOps lifecycle.

- Implemented CI pipeline triggered on commit.
- Configured Continuous Delivery deployment to Azure.
- Managed work using Agile user stories and acceptance criteria traceability.

Advanced Test Automation Frameworks

Playwright Framework:

- Multi-browser execution
- Debug tracing with screenshots
- Payment flow validation
- Flakiness-resistant assertions

Cypress Framework:

- Backend automation
- Docker integration
- Parallel test execution
- Retry logic and error interception
- Video reporting

Puppeteer Framework:

- Accessibility testing
- Device emulation & geolocation testing
- Performance metrics (First Contentful Paint, load time)
- PDF generation automation

ADDITIONAL PROFESSIONAL EXPERIENCE

Real-Time Analyst (PayPal/Venmo)

Sitel Digital Services | 2020–2021

- Managed workforce analytics and queue volume monitoring.
- Generated high-volume operational reports using advanced Excel tools.
- Applied scheduling optimization and shrinkage management principles.
- Strengthened analytical forecasting and performance tracking skills.

Tier 2 Technical Support & Team Supervisor

Teleperformance (Sprint USA / Comcast) | 2014–2017

- Managed escalations for high-value US telecom clients.
- Back-up supervisor for 15-member team.
- Maintained client retention through structured resolution workflows.
- Completed 108-hour leadership certification focused on supervision and emotional intelligence.

VIP Operations Manager & Tour Director

Gran Colombia Tours | 2017–2024

- Designed and led private tour division.
- Managed logistics for high-profile international clients.
- Coordinated large operations including FC Barcelona Legends visit.
- Oversaw quality control and bilingual team execution.

EDUCATION & CERTIFICATIONS

Analysis & Development of Computer Software

QA Automation Engineer

Azure DevOps CI/CD

Software Quality Assurance Bootcamp

Fundamentals of Software Testing

EF SET English Certification – C2